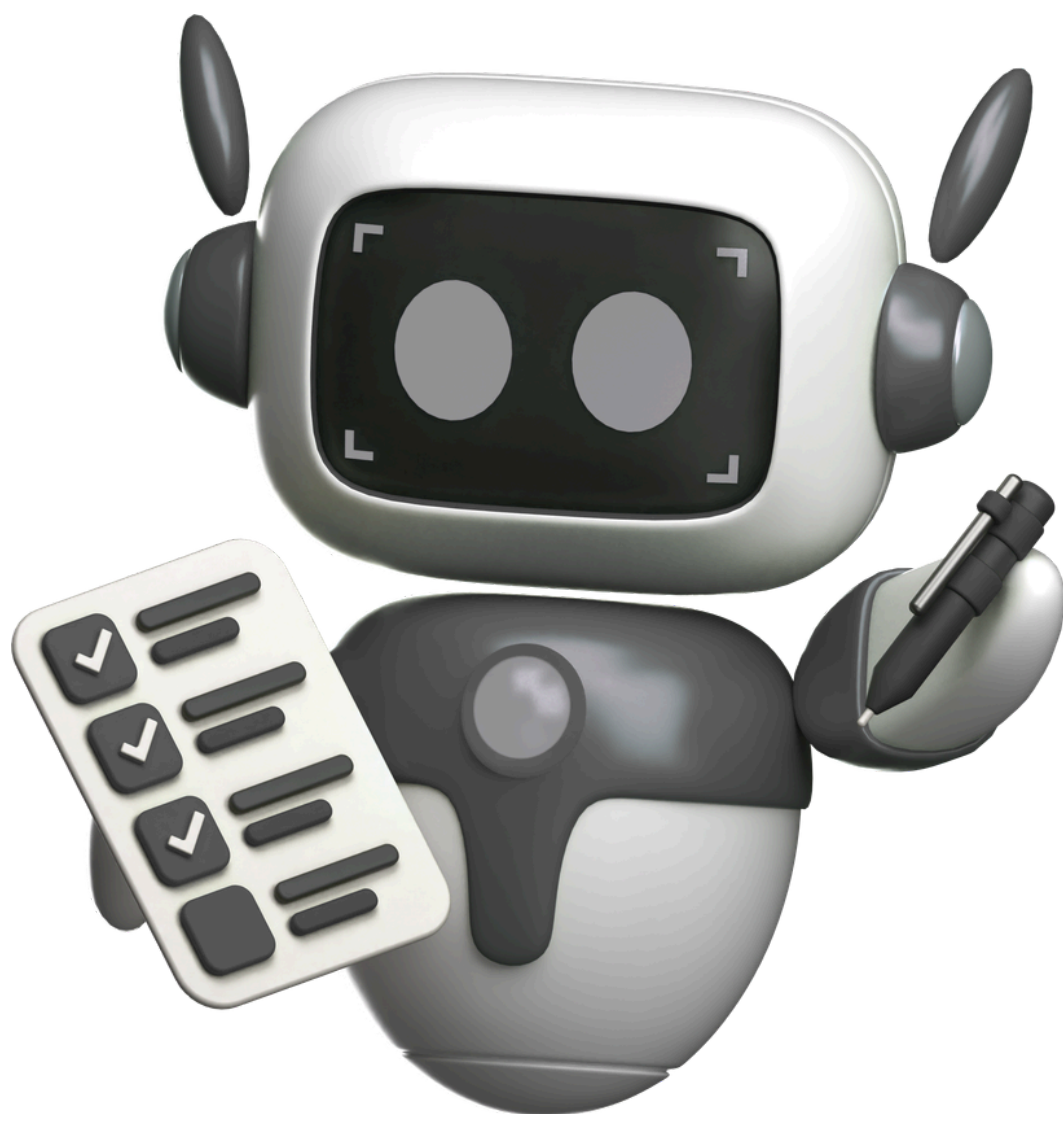
ENCODER VS BI-ENCODER VS CROSS-ENCODER

How retrieval models actually work in RAG systems



HRUTHIK GALI DILEEP

FOLLOW

Repost

What is an Encoder?

An encoder is a neural network that converts text into dense vector representations (embeddings).

What it does:

- Takes text as input
- Processes it to understand context
- Outputs numerical vectors that capture meaning

Example:

Input: "The bank is near the river"
The encoder understands that "bank" = riverbank (not financial institution) based on context.
Output: A vector like [0.23, -0.45, 0.67, ...]

Key point:

Encoders don't generate text. They create representations.
Modern encoders use Transformer architecture (like BERT, Sentence-BERT).

What is a Bi-Encoder?

A bi-encoder uses the same encoder to process query and document separately.

How it works:

Query → Encoder → Query Vector [v1]

Document → Encoder → Document Vector [v2]

Then calculate:

Similarity = cosine(v1, v2)

Example:

Query: "What is RAG?"

→ Vector: [0.8, 0.3, -0.1, ...]

Document: "RAG combines retrieval with generation"

→ Vector: [0.7, 0.4, -0.2, ...]

Cosine similarity = 0.92 (high relevance)

Why it's fast:

Documents are encoded once offline. At query time, only the query needs encoding.

How Bi-Encoder Retrieval Works

Bi-encoders power the first retrieval stage in RAG.

Process:

Offline (one-time):

All documents → Encoder → Store embeddings in vector DB

At query time:

1. User query → Encoder → Query vector
2. Vector similarity search in database
3. Retrieve Top K documents (e.g., Top 100)

Why it scales:

- Documents encoded once
- Search is just vector math (cosine similarity)
- Vector databases (Pinecone, Weaviate) optimize this

Used in almost every production RAG system.

What is a Cross-Encoder?

A cross-encoder processes query and document together in a single pass.

How it works:

[CLS] Query [SEP] Document [SEP] → Encoder → Relevance Score

Example:

Input: "[CLS] What is RAG? [SEP] RAG combines retrieval with generation [SEP]"

The model sees both at once. Attention flows between query and document words.

Output: Direct relevance score (e.g., 0.94)

Key difference:

No separate vectors. The model learns query-document interactions directly.

Trade-off:

Much more accurate. Much slower (can't pre-compute).

Critical limitation:

Not suitable for initial retrieval—you'd need to run it on every query-document pair (millions of combinations).

How Cross-Encoder Re-Ranking Works

After bi-encoder retrieves Top K documents, cross-encoder re-ranks them.

Process:

1. Bi-encoder returns Top 100 documents
2. Take query + each of the 100 documents
3. Pass each pair through cross-encoder
4. Get precise relevance scores
5. Re-rank based on scores
6. Return Top 10 to LLM

Why use this?

Captures nuanced interactions between query and document that bi-encoders miss.

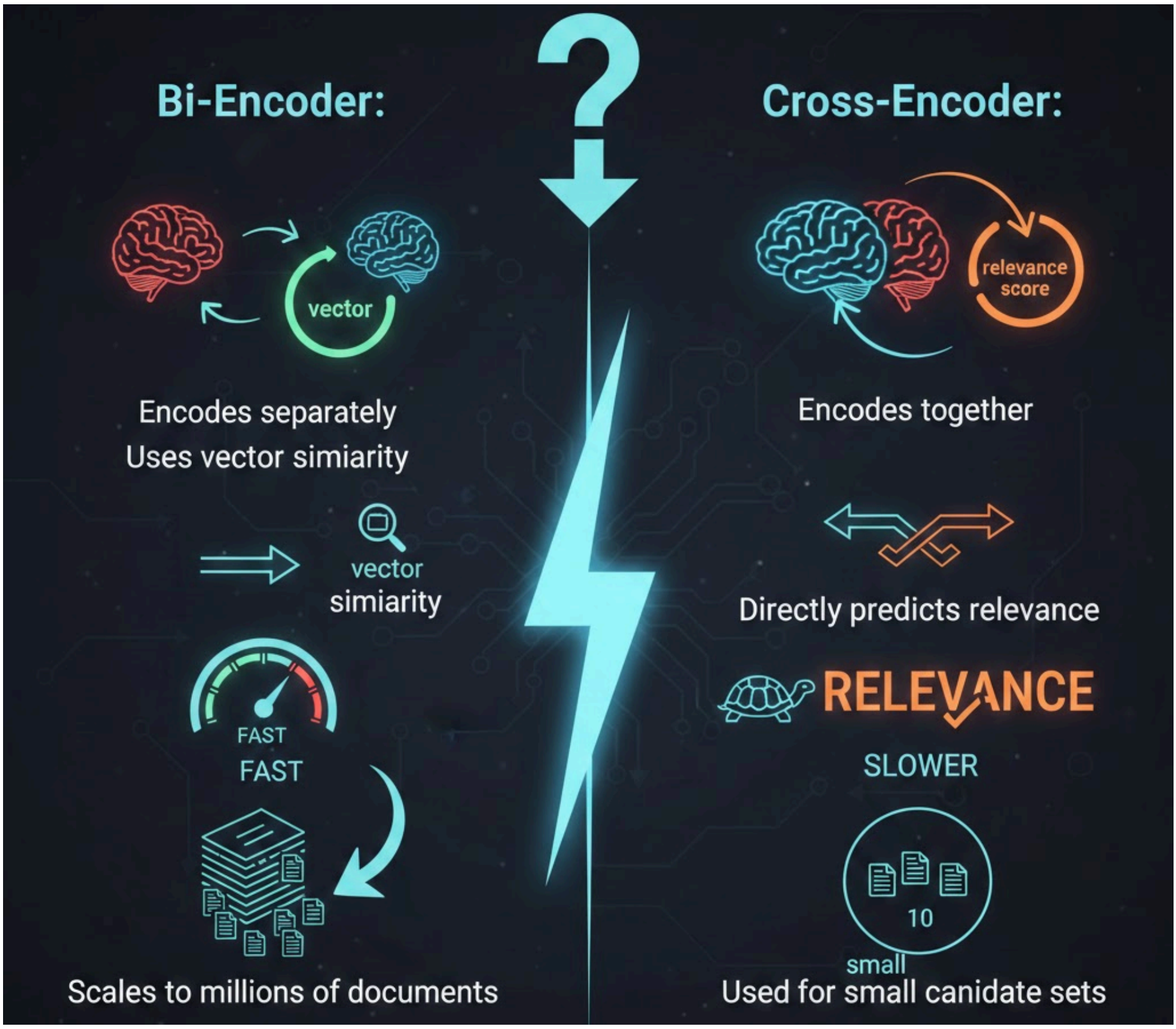
Why not use it alone?

Too slow to run on millions of documents.

Sweet spot:

Re-rank only the top candidates.

Bi-Encoder vs Cross-Encoder



Difference is not the architecture itself.

It’s how inputs are processed.

Final Mental Model

Encoder = Architecture (understands text)

Bi-Encoder = Independent encoding + similarity

Cross-Encoder = Joint encoding + direct scoring

Production RAG pipeline:

Bi-Encoder → Retrieve

Cross-Encoder → Re-rank

That combination gives both speed and accuracy.

